9 **(a)** An electron is travelling in a vacuum towards an electrode with kinetic energy of 8.55×10^{-19} J.

Calculate the stopping potential V_s required to stop the electron.

Loss in KE = Gain in EPE

$$8.55 \times 10^{-19} - 0 = e(V_s - 0) \quad \text{[C1]}$$

$$V_s = \frac{8.55 \times 10^{-19}}{1.60 \times 10^{-19}} = 5.344 \text{ V} = 5.34 \text{ V} \quad \text{[A1]}$$

V_s = ... V [2]

(b) **(i)** The electron in **(a)** is emitted from a material whose work function is 2.80 eV. Calculate the wavelength of the radiation responsible for causing the emission of the electron.

$$\frac{hc}{\lambda} = \phi + KE_{\text{max}}$$

$$\frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{\lambda} = (2.80 \times 1.60 \times 10^{-19}) + 8.55 \times 10^{-19} \quad \text{[C1]}$$

$$\lambda = 1.53 \times 10^{-7} \text{ m} \quad \text{[A1]}$$

wavelength = ... m [2]

(ii) Suggest the type of radiation which has the wavelength in **(b)(i)**.

type of radiation = ...ultraviolet.......... [1]

(c) **(i)** Calculate the de Broglie wavelength of an electron travelling with speed 1.85×10^{7} m s^{-1}.

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

$$\lambda = \frac{6.63 \times 10^{-34}}{(9.11 \times 10^{-31})(1.85 \times 10^{7})} \quad \text{[C1]}$$

$$= 3.93 \times 10^{-11} \text{ m} \quad \text{[A1]}$$

wavelength = ... m [2]

(ii) Graphite, with its layered structure as shown in Fig. 9.1, acts as a natural diffraction grating when used in electron diffraction experiments. The distance between each layer of graphite is 0.335 nm.

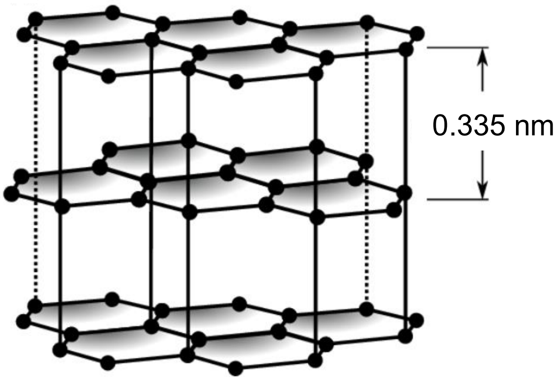

Fig. 9.1

Explain whether electrons having the speed of 1.85×10^{7} m s^{-1} can be used to demonstrate electron diffraction.

The atomic lattice spacing (distance between layers of 3.35×10^{-10} m) is much larger than the de Broglie's wavelength (of 3.93×10^{-11} m), [M1] hence there is negligible/little spreading of electrons. [A1] It is not suitable to demonstrate electron diffraction.

....[2]

(d) Tungsten, a transition metal, is commonly used as a target metal to produce X-rays. The energy levels of the K- to M-shells for tungsten are shown in Fig. 9.2 below.

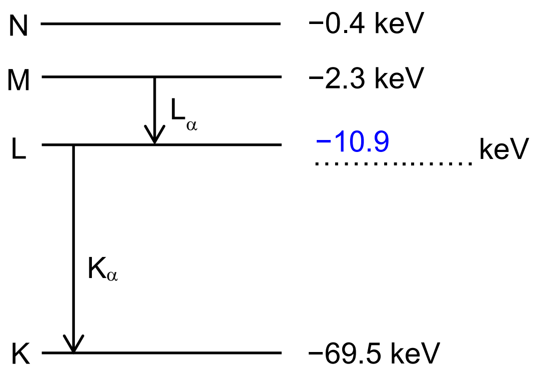

Fig. 9.2 (not to scale)

The wavelength of the photon produced by the K_α transition is 21.2 pm.

(i) Complete Fig. 9.2 by filling in the energy level of the L-shell for tungsten. Show your working clearly.

$$E = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{21.2 \times 10^{-12}} \quad \text{[C1]}$$

$$= 9.382 \times 10^{-15} \text{ J} = 58.64 \text{ keV} \quad \text{[A1]}$$

$$E_2 = -69.5 + 58.64 = -10.9 \text{ keV} \quad \text{[A1]}$$

[3]

(ii) The intensity of various photon wavelengths from electron bombardment of a tungsten target metal is shown in Fig. 9.3. The peak representing K_α transition is labelled.

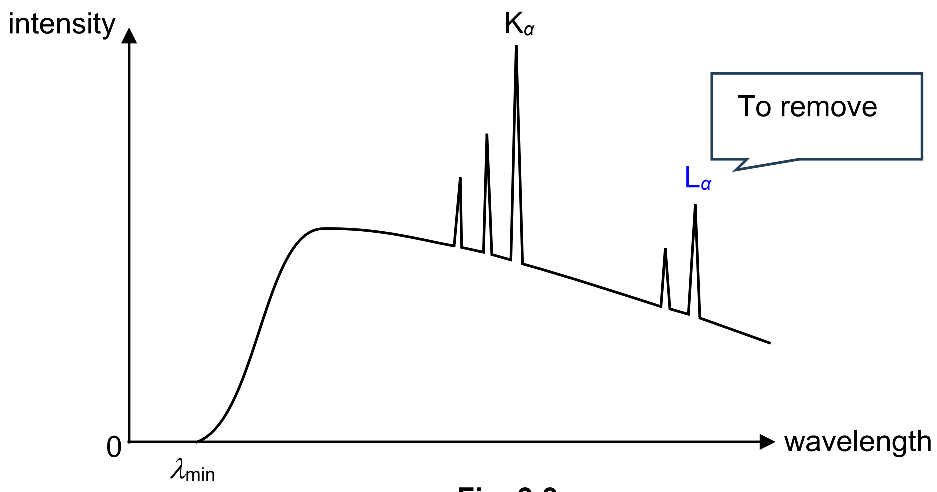

Fig. 9.3

1. On Fig. 9.3, label the peak for L_α transitions. [1]

2. Explain the existence of a minimum wavelength λ_{min}.

The minimum wavelength corresponds to the most energetic X-ray photon produced. [B1]
This happens when a bombarding electron is completely stopped by the target metal and loses all its KE in a single interaction. [B1]

....

....[2]

(iii) With reference to Fig. 9.2, state the minimum energy of the bombarding electrons to produce the characteristic X-rays lines shown in Fig. 9.3.

minimum energy = ...69.5.............. keV [1]

(iv) Explain your answer in **(d)(iii)**.

The bombarding electrons must have sufficient energy to knock out / remove K-shell electron so that orbital electrons from higher level can de-excite to emit photons resulting in characteristic x-ray lines.

....[1]

(e) Fig. 9.4 below shows a typical setup for producing such X-ray beams.

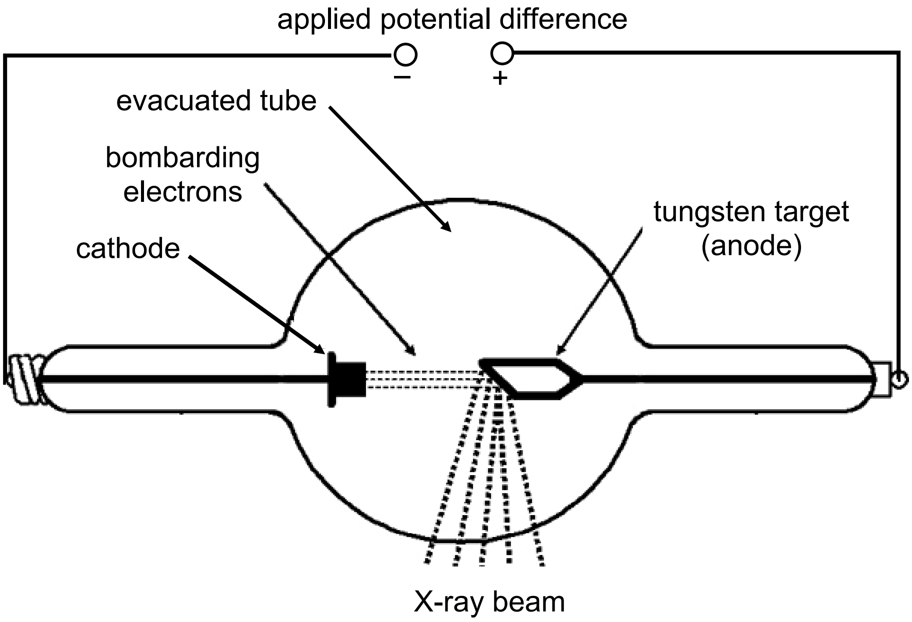

Fig. 9.4

(i) For safety reasons, the wavelength of radiation used for medical X-rays should not be shorter than 50 pm.

Suggest why the wavelength of X-rays radiation should not be shorter than 50 pm.

A shorter wavelength in X-rays translates to higher energy photons, which can be harmful as they can damage living tissue and increase the risk of cancer.

[1]

(ii) Determine the minimum applied potential difference for medical X-rays.

Loss in energy of electron = Maximum photon energy

$q\Delta V = hc/\lambda$

$$\Delta V = \frac{hc}{\lambda q} = \frac{(6.63\times10^{-34})(3.00\times10^{8})}{(50\times10^{-12})(1.60\times10^{-19})}$$ [M1]

$= 24862.5 \text{ V} = 2.5\times10^{4} \text{ V}$ [A1]

minimum potential difference = ... V [2]

[Total: 20]

End of Paper